Kernel Mean Embedding for Causal Inference

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Towards a Learning Theory of Cause-Effect Inference

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ICML - 2015

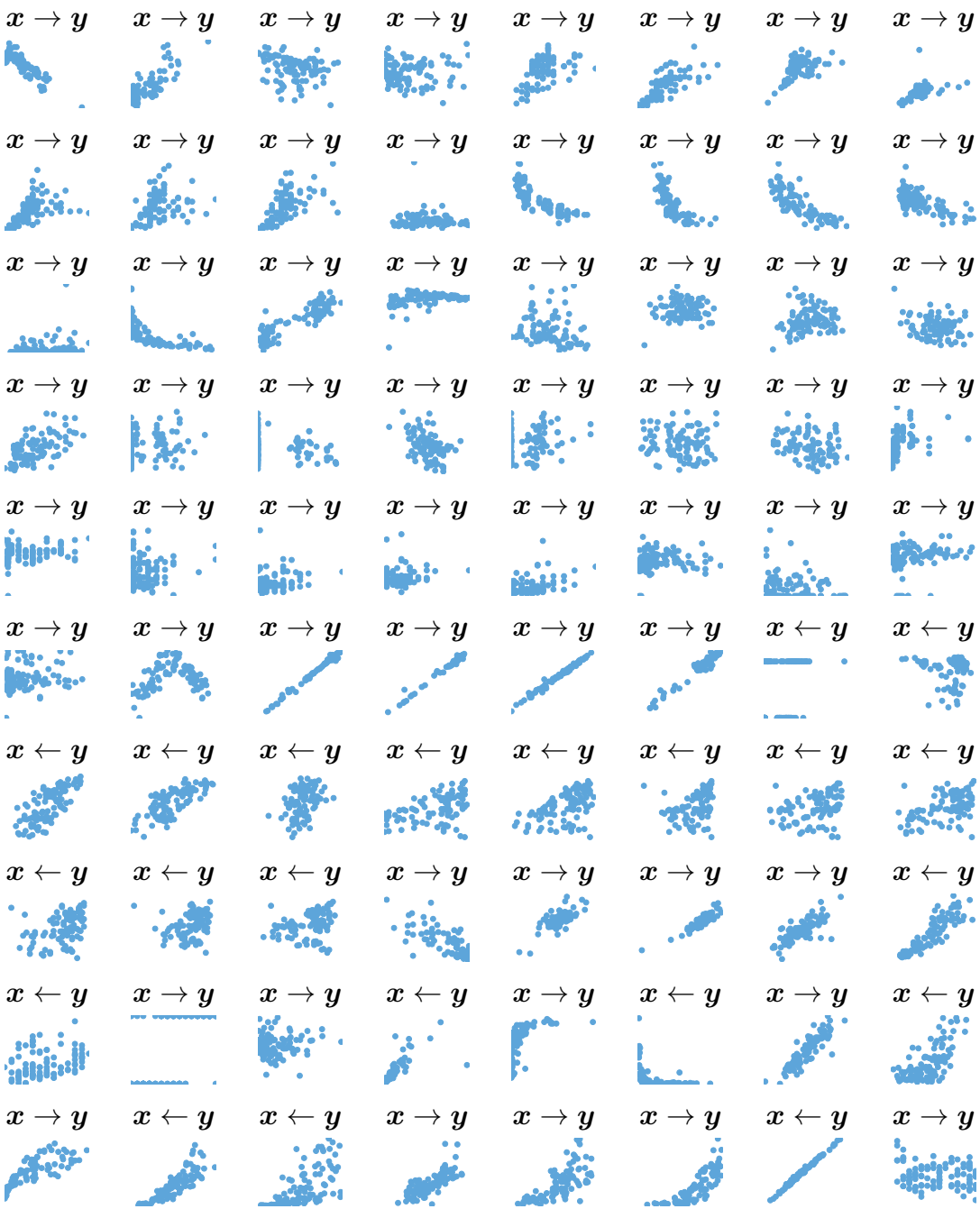
Main Contributions:

- Learning causal relations directly from data.
- Distributional Learning Theory
- Improved Accuracy on MPI Cause Effect Pairs

Main Idea of Paper

Causal inference can been seen as classification of probability distributions. But it is unrealistic to assume have access to the whole distribution. All we know about the distribution is available through iid samples.

$$\left\{ \left(x_i^j \ , \ y_i^j \right) \right\}_{j=1}^{n_i} \sim P_i^{n_i} \longrightarrow \boxed{\text{Classifier}} \longrightarrow x \to y \ \ or \ y \to \ x$$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Kernel Mean Embedding

Definition

Kernel Mean Embedding of the probability distribution P associated with the continous, bounded, positive semi definite kernel $k : \mathcal{Z} \times \mathcal{Z} \to \mathbb{R}$:

$$\mu_k(P) = \int_{\mathcal{Z}} k(z, .)dP = \mathbb{E}_{X \sim P}\Big(k(., X)\Big) \in \mathcal{H}_k \tag{1}$$

Characteristic Kernel

Definition

k is a called a characteristic kernel if:

$$||\mu_k(P) - \mu_k(Q)||_{\mathcal{H}_k} = 0 \iff P = Q \tag{2}$$

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The RBF kernel is an example of characteristic kernels:

$$k(z, z') = \exp(-\gamma||z - z'||_2^2), \quad \gamma > 0$$

This kernel was used in all simulations of the paper.

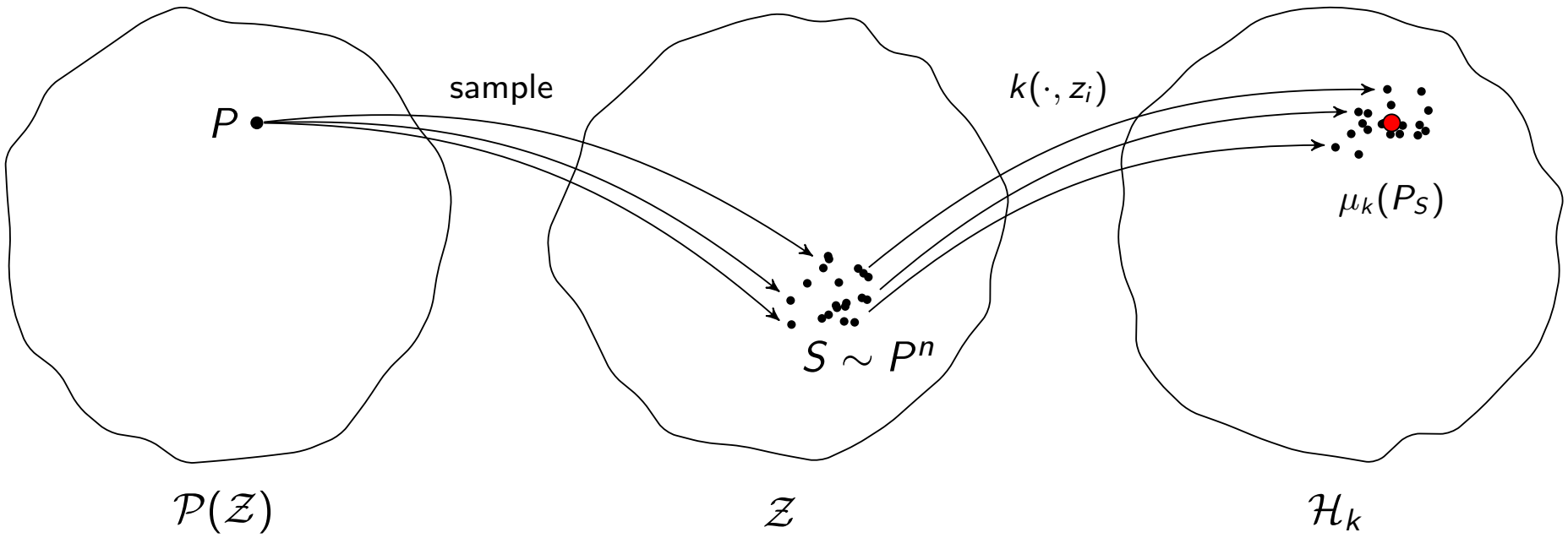

Figure: Transforming a sample S drawn from a distribution P into the empirical mean embedding $\mu_k(P_S)$.

Convergence of Empirical Kernel Mean Embedding

Theorem (Convergence of empirical kernel mean embedding)

Assume that $||f||_\infty \leq 1$ *for* $f \in \mathcal{H}_k$ *with* $||f||_{\mathcal{H}_k} \leq 1$. *Then with probability at least* $1 - \delta$ *we have*

$$||\mu_k(P) - \mu_k(P_S)||_{\mathcal{H}_k} \leq 2\sqrt{\frac{\mathbb{E}_{z\sim P}\left[k(z,z)\right]}{n}} + \sqrt{\frac{2\log(\frac{1}{\delta})}{n}}$$

Proof.

Using the well known dual relation between the norm in RKHS and sup-norm of empirical process, write:

$$||\mu_k(P) - \mu_k(P_S)||_{\mathcal{H}_k} = \sup_{||f||_{\mathcal{H}_k} \leq 1} \langle f, \mu_k(P) - \mu_k(P_S) \rangle =$$

$$\sup_{||f||_{H_k} \leq 1} \left(\mathbb{E}_{z \sim P} f(z) - \frac{1}{n} \sum_{i=1}^{n} f(z_i) \right) = F(z_1, \ldots, z_n)$$

F satisfies the bounded difference condition, thus using McDiarmid's inequality, with $c_i = \frac{2}{n}$ it follows that with probability at least $1 - \delta$:

$$\sup_{\|f\|_{\mathcal{H}_k} \leq 1} \left(\mathbb{E}_{z \sim P} f(z) - \frac{1}{n} \sum_{i=1}^{n} f(z_i) \right)$$

$$\leq \mathbb{E} \sup_{\|f\|_{H_k} \leq 1} \left(\mathbb{E}_{z \sim P} f(z) - \frac{1}{n} \sum_{i=1}^{n} f(z_i) \right) + \sqrt{\frac{2 \log(1/\delta)}{n}}.$$

Some Definitions

For a non-negative cost function $\varphi : \mathbb{R} \to \mathbb{R}^+$ which is surrogate to 0-1 loss, that is $\varphi(z) \geq \mathbb{I}_{z>0}$:

Definition

$$R_\varphi(f) = \mathbb{E}_{(P,l)\sim\mathcal{M}}\Big[\varphi\Big(-f\big(\mu_k(P)\big)l\Big)\Big] \quad f^* = \underset{f\in\mathcal{F}}{\operatorname{argmin}}\, R_\varphi$$

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Definition

$$\hat{R}_\varphi(f) = \frac{1}{n}\sum_{i=1}^{n}\varphi\Big(-f\big(\mu_k(P_i)\big)l_i\Big) \qquad \hat{f}_n = \underset{f\in\mathcal{F}}{\operatorname{argmin}}\, \hat{R}_\varphi$$

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Definition

$$\tilde{R}_\varphi(f) = \frac{1}{n}\sum_{i=1}^{n}\varphi\Big(-f\big(\mu_k(P_{S_i})\big)l_i\Big) \qquad \tilde{f}_n = \underset{f\in\mathcal{F}}{\operatorname{argmin}}\, \tilde{R}_\varphi$$

Main Result of The Paper

Theorem (Excess risk of ERM on empirical kernel mean embeddings)

Consider the RKHS $\mathcal{H}_k$ associated with some bounded, continuous, characteristic kernel function k, such that $\sup_{z\in\mathcal{Z}} k(z,z) \leq 1$. Consider a class $\mathcal{F}$ of functionals mapping $\mathcal{H}_k$ to $\mathbb{R}$ with Lipschitz constants uniformly bounded by $L_{\mathcal{F}}$. Let $\varphi\colon \mathbb{R} \to \mathbb{R}^+$ be a L_φ-Lipschitz function such that $\varphi(z) \geq \mathbb{I}_{z>0}$. Let $\varphi\big(-f(h)l\big) \leq B$ for every $f \in \mathcal{F}_k$, $h \in \mathcal{H}_k$, and $l \in \mathcal{L}$. Then, with probability not less than $1-\delta$ (over all sources of randomness)

$$R_\varphi(\tilde{f}_n) - R_\varphi(f^*) \leq 4L_\varphi R_n(\mathcal{F}_k) + 2B\sqrt{\frac{\log(2/\delta)}{2n}}$$
$$+ \frac{4L_\varphi L_{\mathcal{F}}}{n}\sum_{i=1}^{n}\left(\sqrt{\frac{\mathbb{E}_{z\sim P_i} k(z,z)}{n_i}} + \sqrt{\frac{\log\frac{2n}{\delta}}{2n_i}}\right).$$

Start by decomposing the excess risk as:

$$
\begin{aligned}
R_\varphi(\tilde{f}_n) - R_\varphi(f^*) = & \\
& R_\varphi(\tilde{f}_n) - \tilde{R}_\varphi(\tilde{f}_n) \\
& + \tilde{R}_\varphi(\tilde{f}_n) - \tilde{R}_\varphi(f^*) \\
& + \tilde{R}_\varphi(f^*) - R_\varphi(f^*) \\
& \leq 2 \sup_{f \in \mathcal{F}} |R_\varphi(f) - \tilde{R}_\varphi(f)| \\
& = 2 \sup_{f \in \mathcal{F}} |R_\varphi(f) - \hat{R}_\varphi(f) + \hat{R}_\varphi(f) - \tilde{R}_\varphi(f)| \\
& \leq 2 \sup_{f \in \mathcal{F}} |R_\varphi(f) - \hat{R}_\varphi(f)| + 2 \sup_{f \in \mathcal{F}} |\hat{R}_\varphi(f) - \tilde{R}_\varphi(f)|
\end{aligned}
$$

where $\tilde{R}_\varphi(\tilde{f}_n) - \tilde{R}_\varphi(f^*) \leq 0$.

$$R_\varphi(\tilde{f}_n) - R_\varphi(f^*) \leq 2 \sup_{f \in \mathcal{F}} |R_\varphi(f) - \hat{R}_\varphi(f)| + 2 \sup_{f \in \mathcal{F}} |\hat{R}_\varphi(f) - \tilde{R}_\varphi(f)|$$

The first term has widely been discussed in classic learning theory:

$$\sup_{f \in \mathcal{F}} |R_\varphi(f) - \hat{R}_\varphi(f)| \leq 2L_\varphi \mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{n} \left| \sum_{i=1}^{n} \sigma_i f(z_i) \right| + B\sqrt{\frac{\log(2/\delta)}{2n}}. \quad (3)$$

$$R_\varphi(\tilde{f}_n) - R_\varphi(f^*) \leq 2 \sup_{f \in \mathcal{F}} |R_\varphi(f) - \hat{R}_\varphi(f)| + 2 \sup_{f \in \mathcal{F}} |\hat{R}_\varphi(f) - \tilde{R}_\varphi(f)|$$

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To bound the second term:

$$\begin{aligned}
\sup_{f \in \mathcal{F}} |\hat{R}_\varphi(f) - \tilde{R}_\varphi(f)| &= \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} \Big[\varphi\big(-l_i f\big(\mu_k(P_i)\big)\big) - \varphi\big(-l_i f\big(\mu_k(P_{S_i})\big)\big) \Big] \right| \\
&\leq \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \left| \varphi\big(-l_i f\big(\mu_k(P_i)\big)\big) - \varphi\big(-l_i f\big(\mu_k(P_{S_i})\big)\big) \right| \\
&\leq L_\varphi \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \left| f\big(\mu_k(P_i)\big) - f\big(\mu_k(P_{S_i})\big) \right|,
\end{aligned}$$

$$R_\varphi(\tilde{f}_n) - R_\varphi(f^*) \leq 2 \sup_{f \in \mathcal{F}} |R_\varphi(f) - \hat{R}_\varphi(f)| + 2 \sup_{f \in \mathcal{F}} |\hat{R}_\varphi(f) - \tilde{R}_\varphi(f)|$$

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&\leq \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \left| \varphi\big(-l_i f\big(\mu_k(P_i)\big)\big) - \varphi\big(-l_i f\big(\mu_k(P_{S_i})\big)\big) \right| \\
&\leq L_\varphi \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \left| f\big(\mu_k(P_i)\big) - f\big(\mu_k(P_{S_i})\big) \right|,
\end{aligned}$$

Using the Lipschitzness of f:

$$\sup_{f \in \mathcal{F}} |\hat{R}_\varphi(f) - \tilde{R}_\varphi(f)| \leq L_\varphi \sup_{f \in \mathcal{F}} \frac{L_f}{n} \sum_{i=1}^{n} \|\mu_k(P_i) - \mu_k(P_{S_i})\|_{\mathcal{H}_k}.$$

$$\sup_{f\in\mathcal{F}} |\hat{R}_\varphi(f) - \tilde{R}_\varphi(f)| \leq L_\varphi \sup_{f\in\mathcal{F}} \frac{L_f}{n} \sum_{i=1}^{n} \|\mu_k(P_i) - \mu_k(P_{S_i})\|_{\mathcal{H}_k}. \tag{4}$$

Theorem 1 can be used to upper bound every term in (**??**). We then combine these upper bounds using the union bound over $i = 1, \ldots, n$, and show that for any fixed $P_1, \ldots, P_n$, with probability not less than $1 - \delta$ with respect to the random samples $\{S_i\}_{i=1}^n$, it follows that:

$$\begin{aligned} L_\varphi \sup_{f\in\mathcal{F}} \frac{L_f}{n} \sum_{i=1}^{n} \|\mu_k(P_i) - \mu_k(P_{S_i})\|_{\mathcal{H}_k} \\ \leq L_\varphi \sup_{f\in\mathcal{F}} \frac{L_f}{n} \sum_{i=1}^{n} \left(2\sqrt{\frac{\mathbb{E}_{z\sim P} k(z,z)}{n_i}} + \sqrt{\frac{2\log\frac{n}{\delta}}{n_i}} \right). \end{aligned} \tag{5}$$

$$R_\varphi(\tilde{f}_n) - R_\varphi(f^*) \leq 4L_\varphi \mathbb{E}\sup_{f\in\mathcal{F}} \frac{1}{n}\left|\sum_{i=1}^{n}\sigma_i f(z_i)\right| + 2B\sqrt{\frac{\log(2/\delta)}{2n}}$$
$$+ \frac{4L_\varphi L_{\mathcal{F}}}{n}\sum_{i=1}^{n}\left(\sqrt{\frac{\mathbb{E}_{z\sim P}k(z,z)}{n_i}} + \sqrt{\frac{\log\frac{2n}{\delta}}{2n_i}}\right)$$

where $L_{\mathcal{F}} = \sup_{f\in\mathcal{F}} L_f$.

The typical order of $\mathrm{Rad}(\mathcal{F}_k)$ is $O(n^{-\frac{1}{2}})$. The learning method is consistent as long as $\frac{\log n}{n_i} = o(1)$.

It must be noted that the rate with respect to n_i is not improvable in general as the bound in Theorem 1 is know to be tight.

Extension to The Multivariate Case

d	Number of DAGs with d nodes
1	1
2	3
3	25
4	543
5	29281
6	3781503
7	1138779265
8	783702329343
9	1213442454842881
10	4175098976430598143
11	31603459396418917607425
12	521939651343829405020504063
13	18676600744432035186664816926721
14	1439428141044398334941790719839535103
15	237725265553410354992180218286376719253505
16	83756670773733320287699303047996412235223138303
17	62707921196923889899446452602494921906963551482675201
18	99421195322159515895228914592354524516555026878588305014783
19	332771901227107591736177573311261125883583076258421902583546773505

Empirical Experiments

Throughout their simulations, they featurize the sample $S = \{(x_i, y_i)\}_{i=1}^n$ as

$$\nu(S) = \Big(\mu_{k,m}(P_{S_x}), \mu_{k,m}(P_{S_x}), \mu_{k,m}(P_{S_{xy}})\Big)$$

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- In practice, they set $m = 1000$, and observe no significant improvements when using larger amounts of random features.

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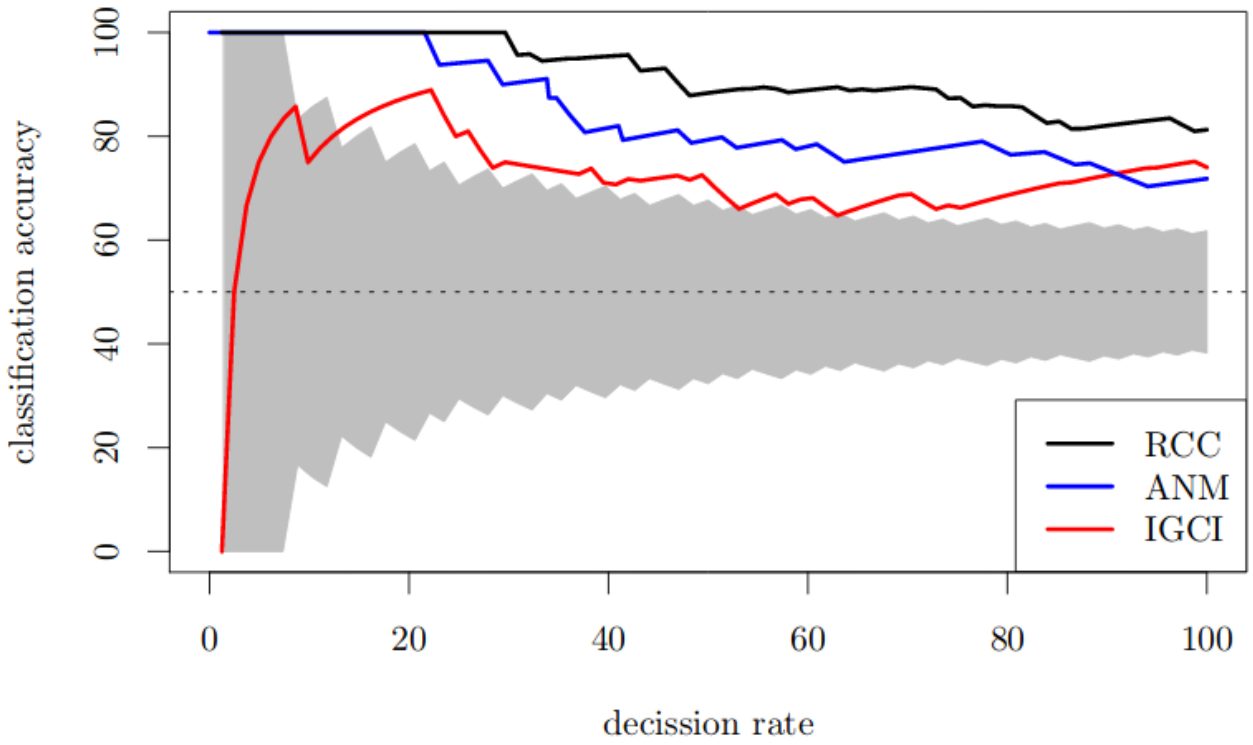
- Motivated by the typical conjecture of existence of asymmetries between the marginal and conditional distributions of causally-related pairs of random variables.
- In practice, they set $m = 1000$, and observe no significant improvements when using larger amounts of random features.
- Random forest is used for classification. The number of trees is chosen from $\{100, 250, 500, 1000, 5000\}$ via cross-validation.

SVM was NOT as good as random foresets

MPI Cause-Effect Pairs

Given the small size of the dataset, they decide to use random artificial data in their training set.



Other Expreiments

- Inferring the arrow of time.
- ChaLearns Challenge Data
  199 Training causal samples S_i, each drawn from the distribution of $X_i \times Y_i$, and labeled either $X_i \to Y_i$, $Y_i \to X_i$, $Xi \leftarrow Zi \to Yi$, or "independent".

Thank you!